Exam 4. Practice 3.

*You may use a basic calculator and formula sheets. Your sheets must contain formulas only, no examples. Phones and all other smart devices are forbidden. Any AI usage will result in an automatic F for the exam. Show **ALL** your work for credits.*

Notice: For all the maximization problems, it is required to do the three-step process to solve the problem.

1. Find $f'(x)$. Simplify 1c.
   a. $f(x) = -2x^3 + 3x + 19$

Using the power rule:

$$f'(x) = 3(-2)x^{3-1} + 1(3)x^{1-1} + 0$$

Answer: $f'(x) = -6x^2 + 3$

b. $f(x) = \frac{2}{x^3} - \frac{4}{x^7} - 5x + 2026$

First, rewrite the function using negative exponents:

$$f(x) = 2x^{-3} - 4x^{-7} - 5x + 2026$$

Apply the power rule:

$$f'(x) = -3(2)x^{-4} - (-7)(4)x^{-8} - 5$$

$$f'(x) = -6x^{-4} + 28x^{-8} - 5$$

c. $f(x) = \frac{3x^3+1}{3x^3-1}$

$$f'(x) = \frac{(9x^2)(3x^3 - 1) - (3x^3 + 1)(9x^2)}{(3x^3 - 1)^2}$$

Factor out $9x^2$ to simplify:

$$f'(x) = \frac{9x^2[(3x^3 - 1) - (3x^3 + 1)]}{(3x^3 - 1)^2}$$

$$f'(x) = \frac{9x^2[3x^3 - 1 - 3x^3 - 1]}{(3x^3 - 1)^2}$$

$$f'(x) = \frac{9x^2(-2)}{(3x^3 - 1)^2}$$

Answer: $f'(x) = -\frac{18x^2}{(3x^3-1)^2}$

d. $f(x) = (2x^2 + 1)(3x^3 + x)$

$$f'(x) = (4x)(3x^3 + x) + (2x^2 + 1)(9x^2 + 1)$$

2.

The per-day cost function (in dollars) for manufacturing a specific high-end electronic component is given by:

$$C(x) = 16,000 + 50x + 2x^2$$

Where x is the number of units manufactured per day ($0 < x \leq 200$).

a. Find the Marginal Cost function.
b. Calculate and interpret the meaning of $C'(40)$.
c. Write the Average Cost function, $AC(x)$.
d. How many items should be manufactured per day in order to minimize the average cost?

a. Find the Marginal Cost function.

The Marginal Cost (MC) is the derivative of the total cost function, $C'(x)$. Using the power rule:

$$C'(x) = (16,000)' + (50x)' + (2x^2)'$$

Answer: $C'(x) = 50 + 4x$

b. Calculate and interpret the meaning of $C'(40)$.

Substitute $x = 40$ into the marginal cost function:

$$C'(40) = 50 + 4(40)$$

$$C'(40) = 50 + 160 = 210$$

Interpretation: At a production level of 40 units per day, the cost is increasing at a rate of **$210 per unit**. This represents the approximate cost of producing the **41st unit**.

c. Write the Average Cost function, $AC(x)$.

To find the Average Cost, divide the total cost by the number of units x:

$$AC(x) = \frac{C(x)}{x} = \frac{16,000 + 50x + 2x^2}{x}$$

Answer: $AC(x) = \frac{16,000}{x} + 50 + 2x$

d. How many items should be manufactured per day in order to minimize the average cost?

Step 1: Find $AC'(x)$

Write the average cost function in power form:

$$AC(x) = 16,000x^{-1} + 50 + 2x$$

Using the power rule:

$$AC'(x) = -16,000x^{-2} + 0 + 2$$

$$AC'(x) = -\frac{16,000}{x^2} + 2$$

Step 2: Solve $AC'(x) = 0$

Set the first derivative to zero to find the value of x:

$$-\frac{16,000}{x^2} + 2 = 0$$

$$2 = \frac{16,000}{x^2}$$

$$2x^2 = 16,000$$

$$x^2 = 8,000$$

$$x = \sqrt{8,000} \approx 89.44$$

Step 3: Verify that $AC''(x) > 0$

Find the second derivative to confirm that $x \approx 89.44$ results in a minimum:

$$AC''(x) = (-2)(-16,000)x^{-3}$$

$$AC''(x) = \frac{32,000}{x^3}$$

Since x is a positive number of units, $\frac{32,000}{x^3}$ results in a value greater than zero ($AC''(x) > 0$). This confirms that the average cost is at its **minimum**.

Answer: To minimize the average cost, approximately **89 items** should be manufactured per day.

3.

A professional pickleball paddle manufacturer has determined that the daily price p (in dollars) they can charge is related to the quantity x of paddles they produce by the following quadratic demand function:

$$p(x) = -x^2 + 60x + 1,500$$

a. Find the revenue function $R(x)$ and the marginal revenue $MR(x)$ function.

b. Interpret the meaning of $MR(10)$.

c. Find the quantity x that should be produced in order to maximize the revenue function.

a. Find the revenue function $R(x)$ and the marginal revenue $MR(x)$ function.

The **Revenue function** is found by multiplying the price $p(x)$ by the quantity x:

$$R(x) = x \cdot p(x) = x(-x^2 + 60x + 1,500)$$

Answer: $R(x) = -x^3 + 60x^2 + 1,500x$

The **Marginal Revenue function** is the derivative of the revenue function, $R'(x)$:
Answer: $MR(x) = -3x^2 + 120x + 1,500$

b. Interpret the meaning of $MR(10)$.

First, calculate the value:

$$MR(10) = -3(10)^2 + 120(10) + 1,500$$

$$MR(10) = -3(100) + 1,200 + 1,500$$

$$MR(10) = -300 + 1,200 + 1,500 = 2,400$$

Interpretation: When 10 paddles are produced and sold per day, the revenue is increasing at a rate of **$2,400 per unit**. This represents the approximate additional revenue generated by selling the **11th paddle**.

c. Find the quantity x that should be produced in order to maximize the revenue function.

Step 1: Find $R'(x)$

(From part a):

$$R'(x) = -3x^2 + 120x + 1,500$$

Step 2: Solve $R'(x) = 0$

Set the marginal revenue to zero to find the value of x:

$$-3x^2 + 120x + 1,500 = 0$$

Divide by -3 to simplify:

$$x^2 - 40x - 500 = 0$$

Factor the quadratic:

$$(x - 50)(x + 10) = 0$$

Since production must be positive, $x = 50$.

Step 3: Verify that $R''(x) < 0$

Find the second derivative to confirm that $x = 50$ results in a maximum:

$$R''(x) = -6x + 120$$

Substitute $x = 50$:

$$R''(50) = -6(50) + 120 = -180$$

Since $-180 < 0$, it confirms that the revenue is at its **maximum** when **50 paddles** are produced per day.

4.

A company determines that when x units of a product are produced each month, they will be sold at a price of:

$$p = 40 - 2x$$

dollars per unit. The total cost of producing x units is:

$$C(x) = x^2 + 10x + 50$$

a. Find the profit function $P(x)$ and the marginal profit $MP(x)$ function.
b. Interpret the meaning of $MP(4)$.
c. How many units should the company produce to maximize the profit?

a. Find the profit function $P(x)$ and the marginal profit $MP(x)$ function.

First, find the **Revenue function**, $R(x)$:

$$R(x) = x \cdot p(x) = x(40 - 2x) = 40x - 2x^2$$

The **Profit function**, $P(x)$, is Revenue minus Cost:

$$P(x) = R(x) - C(x)$$

$$P(x) = (40x - 2x^2) - (x^2 + 10x + 50)$$

$$P(x) = 40x - 2x^2 - x^2 - 10x - 50$$

Answer: $P(x) = -3x^2 + 30x - 50$

The **Marginal Profit function** is the derivative of the profit function, $P'(x)$:**Answer:** $MP(x) = -6x + 30$

b. Interpret the meaning of $MP(4)$.

First, calculate the value:

$$MP(4) = -6(4) + 30$$

$$MP(4) = -24 + 30 = 6$$

Interpretation: At a production level of 4 units per month, the profit is increasing at a rate of **$6 per unit**. This represents the approximate additional profit gained by producing and selling the **5th unit**.

c. How many units should the company produce to maximize the profit?

Step 1: Find $P'(x)$
(From part a):

$$P'(x) = -6x + 30$$

Step 2: Solve $P'(x) = 0$
Set the marginal profit to zero to find the value of x:

$$-6x + 30 = 0$$

$$6x = 30$$

$x = 5$

Step 3: Verify that $P''(x) < 0$
Find the second derivative to confirm that $x = 5$ results in a maximum:

$$P''(x) = -6$$

Since $-6 < 0$, it confirms that the profit is at its **maximum** when **5 units** are produced each month.